

# Which applications should receive review?

A decision-support exercise: rank simulated 90-day default risk while respecting review capacity.

## The decision

Flag a limited set of applications for additional review, not automatic loan rejection.

Inputs: synthetic loan features; target: default within 90 days.

## What counts as useful?

Simulated cost =  $10 \times \text{false negatives} + 1 \times \text{false positives}$ .

Capacity: at most 12% of applications flagged per validation period.

No real-world credit or legal claim.

METHOD

# A leakage-aware, reproducible pipeline

Selection evidence is separated from calibration and from the unlabeled challenge batch.

## DATA & SPLITS

Synthetic train/challenge roles. Forward-time, customer-separated folds; respect label maturity.

## TRAIN & SELECT

Drop leakage features (days\_past\_due\_60, collection\_calls). Fit preprocessing and tuning on eligible training rows only.

## FREEZE & APPLY

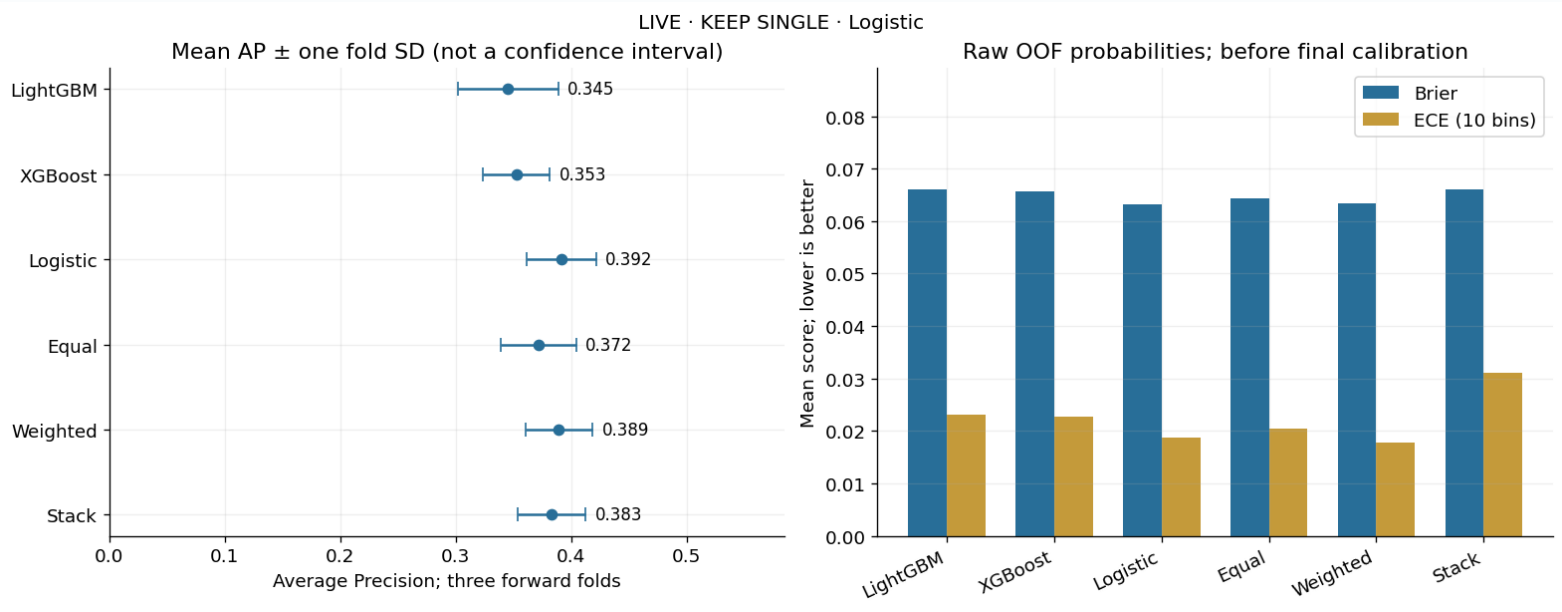
2,155 LIVE OOF rows across three folds support the final comparison. Freeze model, calibrator and policy before challenge scoring.

**Data → forward OOF comparison → KEEP SINGLE → calibration → frozen threshold & 12% cap**

EVIDENCE & ENSEMBLE DECISION

# KEEP SINGLE · Logistic Regression

The predefined Worth-It Gate did not justify additional ensemble complexity.



## Measured decision

Mean OOF AP: Logistic 0.3917;  
Weighted 0.3894; Equal 0.3717.

Fold SD reference: 0.0298 (not a confidence interval).

OOF policy: threshold 0.16892;  
simulated loss 1,111; 245/2,155  
flagged (11.37%).

FN = 10, FP = 1. OOF evidence is not an unbiased challenge performance estimate.

# Explain the model—and the limits

Day 4 interpretation belongs to its Day 4 model; it does not automatically explain final Logistic Regression.

## One Day 4 example

Application TR-009585: raw model probability  $\approx 0.9031$ ; calibrated  $\approx 0.4795$ .

SHAP values were expressed in raw log-odds for the Day 4 tree model; feature contributions are associative, not causal.

## Final calibration diagnostic

On the 836-row calibration-fit sample (78 positives):

Brier: 0.07647 raw  $\rightarrow$  0.07806 sigmoid.

ECE: 0.02112 raw  $\rightarrow$  0.03487 sigmoid.

Neither improved here. In-sample diagnostics are not independent calibration validation.

Further review: independent calibration testing, reliability over time and region-level error stability.

DECISION, LIMITATIONS & FOLLOW-UP

# Freeze, verify, monitor—do not overclaim

The project supports an educational ranking/review workflow, not deployment approval.

## Final operational decision

KEEP SINGLE Logistic Regression.

Frozen OOF threshold: 0.16892 (raw-score policy).

Unlabeled challenge: 2,500 rows; 300 flagged under 12% batch cap. No challenge AP, loss or fairness metric is available.

## What would change my decision?

1. Fresh labeled, time-separated data could change AP, simulated loss or the Worth-It Gate result.
2. Monitor calibration, capacity, temporal drift and region-level errors; retrain/recalibrate only with a new honest validation plan.

Evidence: notebooks 01–05, artifacts/final\_metrics.json, reports, and frozen inference outputs. Repo: dee2002al-c / Advanced-Machine-Learning-Techniques-211--Deema-Abdullah